

H-BRIDGE DRIVER USING PC

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An H-bridge driver circuit is an electronic circuit that allows to control the rotation of motors in either direction by changing the direction of the voltage applied across the load. An H-bridge circuit is often used in robotics and other applications to allow DC motors to run in either forward or backward direction. It can be built using transistors and a few discrete components.

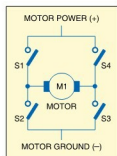


Fig. 1: Basic H-bridge Configuration

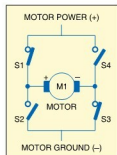


Fig. 2: Clockwise rotation of motor

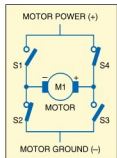


Fig. 3: Anti-clockwise rotation

The direction of applied voltage can easily be controlled with a personal computer (PC) through its parallel port. A similar concept in the form of a simple circuit is presented in this article. This circuit is useful for hobbyists in robotics and computer interfacing.

H-bridge circuit, also called 'full bridge,' has four switching elements placed in the four corners of a rectangle and a motor forming a cross bar, giving it a shape that resembles the 'H' letter. The basic bridge configuration is shown in Fig. 1.

There are four switching

elements in the bridge—S1 (upper side left), S2 (lower side left), S3 (lower side right) and S4 (higher side right). The switches are turned on in pairs, either S1 and S3, or S2 and S4. To power the motor, two diagonally opposed switches have to be turned on. As shown in Fig. 2, when S1 and S3 switches are turned on, the current flows through S1, motor M1 and S3.

As the current flows, the motor begins to rotate in a 'positive' (say, clockwise) direction. Now, if you turn on S4 and S2 switches, the polarity across the motor is reversed (compare the polarities in Fig. 2 and Fig. 3). As the current starts flowing in the opposite direction (S4, M1 and S2), it forces the motor to rotate in reverse direction (say, anti-clockwise).

Parallel port. A standard parallel port interface is a 25-pin, D-type con-



ductor, commonly known as line printer terminal (LPT). It has three different ports, namely, data port, control port and status port. This female connector is normally available on the rear side of a PC. In order to access any port, you should have its port address. The addresses of data port, status port and control port are '0x378', '0x378+1' and '0x378+2', respectively. In this project, two data lines along with signal ground are used for controlling the direction of motor rotation.

Circuit description

Fig. 4 shows the circuit diagram of H-bridge driver using PC. Fig. 6

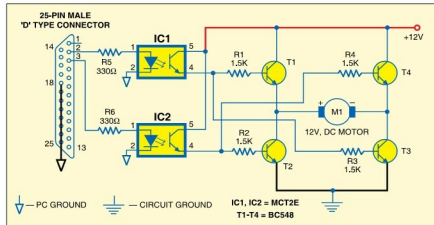


Fig. 4: Circuit diagram of H-bridge driver using PC

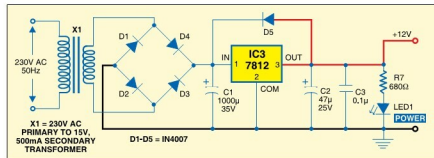


Fig. 5: Power supply circuit